

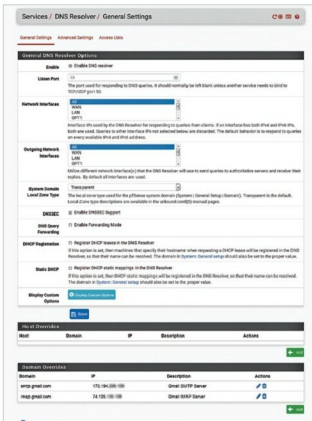


Figure 8: Domain overrides

properly in setting firewall rules.

This can be overcome in two ways:

1. Search all Gmail IP addresses, and allow traffic on GmailServices towards all these IP addresses. This configuration will require considerable effort.
2. Configure **Domain Override** settings under **DNS Resolver** under **Services** to resolve `smtp.gmail.com` and `imap.gmail.com` to correct static IP addresses. Further, configure a rule to allow all traffic for GmailServices towards these IP addresses. Use the following steps:
 1. Find the current IP address belonging to such FQDNs by pinging them.
 2. Configure the DNS resolver to resolve these URLs to these IP address(es).
 3. Use these IP addresses to configure groups and firewall rules.
 4. Ensure that all users use pfSense as their DNS server so that the IP address for overridden domains will resolve to the preconfigured IP address. Note that systems configured with DNS servers other than pfSense will get different IP addresses for these overridden domains and access will be blocked.

Managing firewall configuration changes

As a good change management practice, the following sequence should be observed:

1. Back up the firewall before making any changes – this will enable a rollback of settings if something goes wrong after the change.
2. Make changes in the documentation such as the ACL table first – to ensure the documentation is up to date, always.
3. Make changes to firewall settings, ensuring comments for created rules. Implement only documented rules.
4. Test the change to ensure it is giving only the desired access.
5. Back up the firewall after the change is complete.

Following these steps will ensure correct and up-to-date documentation of implemented firewall rules, which will definitely help future firewall troubleshooting, reconfiguration and migration.

Interesting pfSense features related to firewall rules

pfSense provides easy addition of pass or drop rules by clicking the  signs in the  destination column. Once such a rule is created, do not forget to inspect this rule from **Firewall – Rules – LAN**, and change the default gateway as well as add a proper comment for easy identification at a later date.

Check logs of firewall rule changes from the menu **Diagnostics – Backup and Restore**. Here, details of changed rules can be checked by comparing the earlier rule and changed rule by selecting radio buttons corresponding to the required rules and clicking  on them.

Latest available update

Please update pfSense to the latest new minor release version 2.3.1-RELEASE-p5 which was built on June 16, 2016. Details of new features and changes for this release are available at https://doc.pfsense.org/index.php/2.3.1_New_Features_and_Changes#Update_5. 

References

- [1] TCP Handshake: https://en.wikipedia.org/wiki/Transmission_Control_Protocol
- [2] pfSense firewall rule basics: https://doc.pfsense.org/index.php/Firewall_Rule_Basics
- [3] Example of pfSense basic firewall rule: https://doc.pfsense.org/index.php/Example_basic_configuration
- [4] Note about FQDNs: https://doc.pfsense.org/index.php/Using_FQDNs_in_Aliases
- [5] Forum link for allowing Gmail: <https://forum.pfsense.org/index.php?topic=101809.0>
- [6] URL Alias: <https://forum.pfsense.org/index.php?topic=66499.0>

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